



**ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY  
GUWAHATI**

**Course Structure and Syllabus  
(From Academic Session 2018-19 onwards)**

**B.TECH  
ELECTRICAL AND ELECTRONICS ENGINEERING  
7<sup>th</sup> Semester**



# ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Guwahati

## Course Structure

(From Academic Session 2018-19 onwards)

### B.Tech 7<sup>th</sup> Semester: Electrical and Electronics Engineering

#### Semester VII/ B.TECH/EEE

| Sl. No.                           | Sub-Code    | Subject                          | Hours per Week |   |   | Credit | Marks |     |
|-----------------------------------|-------------|----------------------------------|----------------|---|---|--------|-------|-----|
|                                   |             |                                  | L              | T | P | C      | CE    | ESE |
| Theory                            |             |                                  |                |   |   |        |       |     |
| 1                                 | EEE181701   | Power System Optimisation        | 3              | 1 | 0 | 4      | 30    | 70  |
| 2                                 | EEE1817PE3* | Program Elective-3               | 3              | 0 | 0 | 3      | 30    | 70  |
| 3                                 | EEE1817PE4* | Program Elective-4               | 3              | 0 | 0 | 3      | 30    | 70  |
| 4                                 | EEE1817OE3* | Open Elective-3                  | 3              | 0 | 0 | 3      | 30    | 70  |
| 5                                 | HS181704    | Principles of Management         | 3              | 0 | 0 | 3      | 30    | 70  |
| Practical                         |             |                                  |                |   |   |        |       |     |
| 1                                 | EEE181722   | Project-1                        | 0              | 0 | 6 | 3      | 50    | 50  |
| 2                                 | SI181721    | Internship-III<br>(SAI-Industry) | 0              | 0 | 0 | 2      | -     | 200 |
| TOTAL                             |             |                                  | 15             | 1 | 6 | 21     | 200   | 600 |
| Total Contact Hours per week : 22 |             |                                  |                |   |   |        |       |     |
| Total Credit: 21                  |             |                                  |                |   |   |        |       |     |

| Program Elective-3 |              |                                                                                 |
|--------------------|--------------|---------------------------------------------------------------------------------|
| Sl. No.            | Subject Code | Subject                                                                         |
| 1                  | EEE1817PE31  | Reliability Engineering                                                         |
| 2                  | EE1817PE33   | Pattern Recognition and Machine Learning                                        |
| 3                  | EEE1817PE3*  | Any other subject offered from time to time with the approval of the University |

| <b>Program Elective-4</b> |                     |                                                                                 |
|---------------------------|---------------------|---------------------------------------------------------------------------------|
| <b>Sl. No.</b>            | <b>Subject Code</b> | <b>Subject</b>                                                                  |
| 1                         | EEE1817PE41         | Digital Signal Processing                                                       |
| 2                         | EE1817PE43          | Advanced Electrical Drives                                                      |
| 3                         | EEE1817PE4*         | Any other subject offered from time to time with the approval of the University |

| <b>Open Elective-3</b> |                     |                                                                                 |
|------------------------|---------------------|---------------------------------------------------------------------------------|
| <b>Sl. No.</b>         | <b>Subject Code</b> | <b>Subject</b>                                                                  |
| 1                      | EEE1817OE31         | Illumination Engineering                                                        |
| 2                      | EEE1817OE3*         | Any other subject offered from time to time with the approval of the University |

### Detailed Syllabus:

| Course Code | Course Title              | Hours per week<br>L-T-P | Credit<br>C |
|-------------|---------------------------|-------------------------|-------------|
| EEE181701   | Power System Optimisation | 3-1-0                   | 4           |

#### Course Objectives:

The objective of this course is to provide students with a broad understanding of optimization of a Power system network in its operation and handling faults

#### Course Outcomes:

**CO1:** Economics of Power System network

**CO2:** Make use of various modes of hydro-thermal scheduling

**CO3:** Applying the concept of optimality in Power System networks

**CO4:** Utilize knowledge of Power system state Estimation in power system

#### **MODULE 1: Economic Operation of Power System**

Methods of loading Turbo generator and its modelling, input-output curve, heat rate curve and incremental fuel cost, Economic load dispatch with and without loss (lambda iteration), Loss coefficients, Based point, Participation factor, Economic load dispatch by gradient search method and Newtons method, unit commitment

#### **MODULE 2: Load Flow Analysis**

Introduction, Node elimination by Kron's reduction, Z bus matrix formulation, Y-bus matrix formulation

#### **MODULE 3: Optimal Power Flow (OPF)**

OPF Formulation, Optimization by Newton's method, OPF by (considering equality constraint with Newton's method and penalty function imposed), OPF (considering equality constraint), OPF by Linear programming method, OPF by gradient method

#### **MODULE 4: Optimal Hydrothermal Scheduling**

Hydrothermal scheduling formulation, Short term hydrothermal scheduling, hydrothermal scheduling including loss, Pumped storage hydro plant

#### **MODULE 5: Power System Security**

Introduction, Contingency analysis, Security constrained optimal power flow, Factors affecting power system security, Power system state estimation.

#### **Textbooks/ Reference Books:**

1. Power generation operation and control by Allen J. Wood and Bruce F.Wollenberg
2. Power system optimization by D.P.Kothari and J.S.Dhillon
3. Operation and control in power systems by P.S.Murty

| Course Code | Course Title            | Hours per week<br>L-T-P | Credit<br>C |
|-------------|-------------------------|-------------------------|-------------|
| EEE1817PE31 | Reliability Engineering | 3-0-0                   | 3           |

**Course Objective: Power system Reliability**

**Course outcomes:** At the end of the course the students will be able to:

**CO1:** Modelling Power system based on Reliability scale

**CO2:** Six Sigma Effect

**MODULE 1: Introduction**

Need for Reliability and Quality, Reliability and Quality Terms and Definitions, Performance, Quality, and Reliability, Reliability as a relative measure and its life cycle.

**MODULE 2: Reliability Mathematics**

Estimation of Un-Reliability, Success rate and Failure rate, Mean time to failure, Mean time between the failure, Hazard rate and its curve (Bath tub curve), Conditional Reliability, Total probability theory, Theorem of Independent events , Probability distribution function, Cumulative distribution function, Reliability Network modelling, Binomial distribution, Arithmetic Mean ,Variance, Standard deviation, Markov Modelling, Redundant system.

**MODULE 3: Power System Reliability**

Introduction, Terms and Definitions, Service Performance Indices, Loss of Load Probability, Models for Performing Availability Analysis of a Single Generator Unit, Models for Performing Availability Analysis of Transmission and Associated Systems.

**MODULE 4: Economics of Reliability Engineering**

Reliability design cost-internal and external, Life cycle cost, Reliability achievement cost model, Reliability utility cost model, Depreciation cost model.

**MODULE 5: Six Sigma-Reliability**

Six Sigma Fundamental,Six Sigma Implementation, Steps in the Six Sigma Process, Optimization Problems in the Six Sigma Process.

**Textbooks/ Reference Books:**

1. Reliability,Quality and Safety for Engineering; Author: B.S.Dhillon; Publisher: CRC Press
2. Reliability Engineering; Author: E Balaguruswamy;Publisher: McGraw Hills Education
3. Reliability Engineering; Author:Kailash C. Kapur,Michael Pecht;Publisher:WILEY
4. Applied Reliability and Quality Fundamentals, Methods and Procedures; Author:B.S. Dhillon; Publisher:SPRINGER

| Course Code | Course Title                             | Hours per week<br>L-T-P | Credit<br>C |
|-------------|------------------------------------------|-------------------------|-------------|
| EE1817PE33  | Pattern Recognition and Machine Learning | 3-0-0                   | 3           |

### **COURSE OBJECTIVES:**

To teach concepts of inculcating artificial intelligence in machines by different computational methods

**COURSE OUTCOMES:** At the end of this course, the students will be able to

**CO1:** describe structure, components and mathematical formulation of learning by machines

**CO2:** develop algorithms for learning and pattern recognition by machines

**CO3:** analyze performances and characteristics of machine learning and pattern recognition algorithms for utilization in societal, academic and industrial purposes

### **MODULE 1: Introduction to Machine Learning**

Intelligent Machines, Data Representation, Diversity of Data, Forms of learning, Machine learning and data mining, Linear Algebra and Machine Learning Techniques

### **MODULE 2: Supervised Learning**

Learning from observation, Bias and Variance, Computational Learning Theory, Estimating Generalization errors, Metrics for assessing regression, Metrics for assessing classification

### **MODULE 3: Statistical Learning**

Descriptive Statistics in learning Techniques, Bayesian Reasoning, k-Nearest Neighbour (kNN) Classifier, Discriminant functions and Regression functions, Linear Regression with Least Square Error criterion, Logistic Regression for Classification Tasks

### **MODULE 4: Learning with Support Vector Machine (SVM)**

Linear Discriminant Function for binary classification, Linear Maximal Margin classifier for overlapping classes, Non Linear classifier, Regression by SVM, Variants of basic SVM Techniques

### **MODULE 5: Learning with Neural Networks**

Cognitive Machines, Neuron Models, Network Architectures, Perceptrons, Linear Neurons, Error correction delta Rule, Multilayer perceptron (MLP) networks, Radial Basis Function (RBF) networks, Genetic Neural Systems

### **MODULE 6: Fuzzy Inference System**

Cognitive Uncertainty and Fuzzy Rule Base, Fuzzy Quantification of Knowledge, Fuzzy Rule Base and Approximate Reasoning, Mamdani Model, Neuro Fuzzy Inference model, Genetic Fuzzy Systems

### **MODULE 7: Data Clustering and Data Transformation**

Unsupervised Learning, Engineering Data, Overview of basic clustering models, K-Means Clustering, Fuzzy k-means clustering, some useful data transformation, Introduction to PCA, Decision Tree Learning, Measuring of Impurity for evaluating Splits in Decision Trees, Pruning Decision Trees, Strengths and weaknesses of Decision Tree Approach

### **MODULE 8: Application of Machine Learning**

Some Practical Application Implementation

### **Textbooks/ Reference Books:**

1. M Gopal: Applied Machine Learning, McGraw Hill Education
2. Geoffrey J. McLachlan: Discriminant Analysis and Statistical Pattern Recognition, John Wiley & Sons
3. Richard O. Duda, Peter E. Hart, David G. Stork: Pattern Classification, John Wiley & Sons
4. S. Theodoridis, K. Koutroumbas: Pattern Recognition, Elsevier
5. Keinosuke Fukunaga: Introduction to statistical pattern recognition, Morgan Kaufmann, Academic Press
6. Shai Shalev-Shwartz, Shai Ben-David: Understanding Machine Learning: From Theory to Algorithms, Cambridge University Press, 2015
7. S. Rajasekaran, G. A. Vijayalakshmi Pai: Neural Networks, Fuzzy Systems and Evolutionary Algorithms: Synthesis and Applications, PHI 2017.
8. B. Yegnanarayana: Artificial Neural Networks, PHI 2015
9. Simon Haykin: Neural Networks, Pearson Education, 2003.
10. Laurance Fausett: Fundamentals of Neural Networks, Englewood cliffs, N.J., Pearson Education, 1992.
11. Jacek M. Zurada: Introduction to Artificial Neural Systems, Jaico Publishing Home, 2002.
12. U Dinesh Kumar Manaranjan Pradhan: Machine Learning using Python, John Wiley & Sons 2019
13. B. Kosko: Neural Networks and Fuzzy Systems, Prentice-Hall of India Pvt. Ltd., 1994.
14. G. J. Klir and T. A. Folger: Fuzzy Sets, Uncertainty and Information, Prentice-Hall of India Pvt. Ltd., 1993.
15. H.J. Zimmermann: Fuzzy Set Theory and Applications, Allied Publication Ltd., 1996.
16. Timothy J. Ross: Fuzzy Logic with Engineering Applications, Tata McGraw Hill, 1997
17. John Yen & Reza Langari: Fuzzy Logic – Intelligence Control & Information, Pearson Education, New Delhi, 2003.
18. Driankov, Hellendroon: Introduction to Fuzzy Control, Narosa Publishers.
19. David Goldberg: Genetic Algorithms and Machine learning, PHI
20. Z Michalewicz: Genetic Algorithms + Data Structures = Evolution Programs, 3rdEd, Springer, 1996.
21. T Baeck, D B Fogel, Z Michalewicz: Evolutionary Computation Vol 2 Advanced Algorithms and Operators, Institute of Physics Publishing, Bristol, UK, 2000
22. Y. Gong, W. Xu: Machine Learning for Multimedia Content Analysis, Springer
23. Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar: Foundations of Machine Learning, MIT Press.
24. Simon Haykin: Neural Networks & Learning Machines, Pearson 2016

| Course Code | Course Title              | Hours per week<br>L-T-P | Credit<br>C |
|-------------|---------------------------|-------------------------|-------------|
| EEE1817PE41 | Digital Signal Processing | 3-0-0                   | 3           |

**Course Outcome:** At the end of this course students will be able to

**CO1:** Represent signals mathematically in continuous and discrete time and frequency domain.

**CO2:** Apply z-transform for the analysis of signals and LSI systems.

**CO3:** Deduce filter structures.

**CO4:** Apply Fast Fourier transform techniques to derive discrete Fourier transformation of signals and systems.

**CO5:** Design of different types of digital filters for various applications

**CO6:** Explain quantization errors and their effects on performance of digital signal processing applications and estimate spectral density of random signals.

### **MODULE 1: Introduction**

What is DSP, block diagram of DSP System, its application and advantages, Sampling Theorem  
Overview of Discrete time Systems: Definition, representation, impulse response, derivation of the output sequence, linear convolution, graphical and analytical method, stability and causality condition, FIR and IIR systems. Linear Time Invariant (LTI) systems characterized by constant coefficient difference equations. Inverse System

### **MODULE 2: Analysis of LSI systems using Z Transform**

Transfer function of LSI discrete time system, convolution and deconvolution using z transform, causality and stability of LSI discrete time system

### **MODULE 3: Realization of Digital Systems**

Recursive and non-recursive structures. Block diagram and signal flow graphs. Cascade and parallel realizations

### **MODULE 4: Discrete Fourier Transforms**

Discrete Fourier transform (DFT) and properties. Linear and Circular convolution. Computation of DFT. Fast Fourier Transforms (FFT) -basic D-I-T and D-I-F algorithms

### **MODULE 5: Design of Digital Filters**

IIR Filter Design: Design of Lowpass, Bandpass, Bandstop and High pass filter using Butterworth, Chebyshev, FIR filters Design: Window method

### **MODULE 6: Finite Word Length Effects**

Effect of finite register length in FIR filter design, Quantization errors and their effects on performance of digital signal processor



## **MODULE 7: Power Spectral Approximation**

Parametric and non-parametric spectral estimation

## **MODULE 8: Multi-Rate Signal Processing**

Introduction to multi-rate signal processing

### **Textbooks/Reference Books:**

1. S. K. Mitra, “Digital Signal Processing: A computer based approach”, McGraw Hill, 2011.
2. A.V. Oppenheim and R. W. Schaffer, “Discrete Time Signal Processing”, Prentice Hall, 1989.
3. J. G. Proakis and D.G. Manolakis, “Digital Signal Processing: Principles, Algorithms And Applications”, Prentice Hall, 1997.
4. L. R. Rabiner and B. Gold, “Theory and Application of Digital Signal Processing”, Prentice Hall, 1992.
5. J. R. Johnson, “Introduction to Digital Signal Processing”, Prentice Hall, 1992.
6. D. J. DeFatta, J. G. Lucas and W. S. Hodgkiss, “Digital Signal Processing”, John Wiley & Sons, 1988.
7. A. Nagoorkani, “Digital Signal Processing”, McGraw Hill Education (India) Pvt. Ltd (2e)
8. J.G. Proakis, D.G. Manolakis and D. Sharma, “Digital Signal Processing: Principles, Algorithm and Application”, Pearson
9. P. Ramesh Babu, “Digital Signal Processing”, Scitech

| Course Code | Course Title               | Hours per week<br>L-T-P | Credit<br>C |
|-------------|----------------------------|-------------------------|-------------|
| EE1817PE43  | Advanced Electrical Drives | 3-0-0                   | 3           |

**Course Outcomes:** At the end of this course the student will be able to:

1. Analyze the operation of power electronic converters and their control strategies
2. Analyze the vector control strategies for ac motor drives
3. Implement efficient control strategies for ac motor drives

#### **MODULE 1: Power Converters for AC drives**

PWM control of inverter, selected harmonic elimination, space vector modulation, current control of VSI, three level inverter, Different topologies, SVM for 3 level inverter, Diode rectifier with boost chopper, PWM converter as line side rectifier, current fed inverters with self-commutated devices. Control of CSI, H bridge as a 4-Q drive

#### **MODULE 2: Induction Motor Drives**

Different transformations and reference frame theory, modeling of induction machines, voltage fed inverter control-v/f control, vector control, direct torque and flux control (DTC)

#### **MODULE 3: Synchronous Motor Drives**

Modeling of synchronous machines, open loop v/f control, vector control, direct torque control, CSI fed synchronous motor drives

#### **MODULE 4: Permanent Magnet Motor Drives**

Introduction to various PM motors, BLDC and PMSM drive configuration, comparison, block diagrams, Speed and torque control in BLDC and PMSM

#### **MODULE 5: Switched Reluctance Motor Drives**

Evolution of switched reluctance motors, various topologies for SRM drives, comparison, Closed loop speed and torque control of SRM

#### **MODULE 6: DSP Based Motion Control**

Use of DSPs in motion control, various DSPs available, realization of some basic blocks in DSP for implementation of DSP based motion control

#### **Textbooks/ Reference Books:**

1. P. C. Krause, O. Wasynczuk and S. D. Sudhoff: Analysis of Electric Machinery and Drive Systems, John Wiley & Sons, 2013
2. H. A. Taliyat and S. G. Campbell: DSP based Electromechanical Motion Control, CRC press, 2003
3. R. Krishnan: Permanent Magnet Synchronous and Brushless DC motor Drives, CRC Press, 2009
4. B. K. Bose: Modern Power Electronics and AC Drives, Pearson Education, Asia, 2003

| Course Code | Course Title             | Hours per week<br>L-T-P | Credit<br>C |
|-------------|--------------------------|-------------------------|-------------|
| EEE1817OE31 | Illumination Engineering | 3-0-0                   | 3           |

### **COURSE OBJECTIVES**

- This course is an introduction to the fundamental concepts and techniques in lighting design and their applications to solve real life problems

**COURSE OUTCOMES:** After completion of the course students will be able to

**CO1:** Lighting design calculations

**CO2:** Software based evaluation

### **MODULE 1: Illumination Basics:**

Light, Light spectrum, Solid angle, Photometric and Radiometric definitions and units

### **MODULE 2: Concepts of Optics:**

Reflection, Refraction, Transmission, Absorption, Diffusion, Eye and vision

### **MODULE 3: Illumination Laws:**

Inverse square law, Lambert cosine law, Lambertian Emission and Reflection law

### **MODULE 4: Lighting Schemes:**

Direct lighting, Semi direct lighting, Semi Indirect lighting, Indirect lighting, General diffusion lighting

### **MODULE 5: Lighting Calculation:**

Watt per square method, Lumen or Light flux method, Point to point or inverse square law method, illumination calculation

### **MODULE 6: Artificial Light Sources:**

Incandescent lamp, Fluorescent lamp, Mercury vapour lamp, Sodium vapour lamp

### **Textbooks/ Reference Books:**

1. Book1: Illuminating engineering by Kamallesh Roy (Laxmi Publications Pvt ltd.)
2. Web Link: NPTEL (National programme on Technology enhanced learning)

| Course Code | Course Title             | Hours per week<br>L-T-P | Credit<br>C |
|-------------|--------------------------|-------------------------|-------------|
| HS181704    | Principles of Management | 3-0-0                   | 3           |

#### **MODULE1: Introduction**

Definition and meaning of management, Characteristics of management, importance of management, functions of management-planning, organising, directing, staffing, coordination and controlling etc., principles of management, Difference between administration and management

#### **MODULE2: Financial Management**

Definition and management of financial planning, importance and characteristics of sound financial plan, concepts of capital- fixed capital and working capital, source of finance, fund flow statement.

#### **MODULE3: Marginal costing**

Definition and meaning of marginal costing, advantages, marginal cost equation, contribution, profit-volume ratio, break even analysis, margin of safety.

#### **MODULE4: Cost Accounting**

Cost Accounting- Concept and benefit, elements of cost, preparation of cost sheet with adjustment of raw materials, work-in-progress and finished goods.

#### **MODULE5: Capitalisation**

Definition and meaning of capitalisation, over and under capitalisation.

#### **MODULE6: Motivation**

Introductory observation, definition of motivation, motivational technique, features of sound motivational system.

#### **MODULE7: Leadership**

Concept of leadership, principles of leadership, functions of leadership, qualities of leadership, different styles of leadership.

#### **Textbooks/Reference Books:**

1. Principle of Business Management: RK Sharma, Shashi K.Gupta
2. Business Organisation and Management: SS Sarkar, RK Sharma, Shashi K.Gupta
3. Industrial Organisation and Management: SK Basu, KC Sahu, B Rajvive
4. Principles of Management by Dr. A. K. Bora: Chandra Prakash, Guwahati.
5. Management Accounting: RK Sharma, Shashi K Gupta
6. Cost Accounting: SP Jain, K I Narang
7. Cost Accounting, RSN Pillai, V Bhagawati
8. Principles of Management: RN Gupta
9. Principles of Management: RSN Pillai, S. Kala
10. Principles of Management: Dipak Kumar Bhattacharj

| <b>Course Code</b>                                                   | <b>Course Title</b> | <b>Hours per week<br/>L-T-P</b> | <b>Credit<br/>C</b> |
|----------------------------------------------------------------------|---------------------|---------------------------------|---------------------|
| <b>EEE181722</b>                                                     | <b>Project-1</b>    | <b>0-0-6</b>                    | <b>3</b>            |
| <b>GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME</b> |                     |                                 |                     |

| <b>Course Code</b>                                                   | <b>Course Title</b>                        | <b>Hours per week<br/>L-T-P</b> | <b>Credit<br/>C</b> |
|----------------------------------------------------------------------|--------------------------------------------|---------------------------------|---------------------|
| <b>SI181721</b>                                                      | <b>Internship-III<br/>(SAI - Industry)</b> | <b>0-0-0</b>                    | <b>2</b>            |
| <b>GUIDELINES WILL BE ISSUED BY THE UNIVERSITY FROM TIME TO TIME</b> |                                            |                                 |                     |

\*\*\*\*\*